

PROJECT REPORT

Heavy Equipment Selling Price Prediction Challenge

Predicting Heavy Equipment Resale Value with a Tuned Gradient-Boosted Regression Pipeline

HEAVY_EQUIPMENT_SELLING_PRICE_PREDICTION | MLP DIPLOMA PROJECT SUBMISSION
Challenge: Heavy Equipment Selling Price Prediction Challenge — Regression (MLP Project, 2026 T2)

Abstract

Forecasting the resale price of heavy industrial equipment from transactional and technical specification records is a practically important regression problem for secondary-market pricing and asset valuation. This project builds a complete, disciplined regression pipeline that predicts **TargetValue** — the transaction price of a piece of heavy equipment — from operational metadata, geographic indicators, and a large block of anonymized technical specification fields, evaluated using Root Mean Squared Logarithmic Error (RMSLE). Following exploratory analysis of target skew, missingness patterns, and categorical cardinality, the raw fields were denoised and combined with engineered features — equipment age, usage intensity, parsed specification tokens, and a missing-spec count — into a 55-feature model-ready dataset. Five candidate regressors — Ridge Regression, Random Forest, XGBoost, LightGBM, and CatBoost — were trained and compared on a common validation split, with the strongest model (XGBoost) then refined via randomized hyperparameter search. The tuned model achieved a validation RMSLE of 0.2066, closely matching a leaderboard score of 0.20491, and the final model was refit on the full training set before generating the competition submission.

1. Introduction

1.1 Problem Statement

Estimating what a used piece of heavy equipment — excavators, loaders, and similar industrial machinery — will sell for is a high-value regression problem for resale marketplaces, lenders, and fleet operators. The task is framed as a supervised regression problem: given transactional metadata (transaction date, region, vendor), operational indicators (manufacture year, meter hours, utilization tier), and a large block of technical specification fields, predict the continuous target **TargetValue**. The dataset combines a small number of clean numeric fields with a much larger set of categorical and high-cardinality technical descriptor fields, many of which carry substantial and structurally patterned missingness, all of which must be understood and handled deliberately before reliable modeling can begin.

1.2 Objectives

- Profile the dataset to identify skew in the target variable, missingness patterns, and the numeric/categorical balance of features.

- Recover usable signal from high-cardinality and free-text fields (transaction date, full specification descriptor).
- Engineer age-, usage-, and specification-derived features that capture non-obvious price drivers.
- Train and compare multiple regression models — linear and tree-based — on a common validation split.
- Select and tune a final model based on RMSLE, the competition's scoring metric.
- Generate a correctly formatted submission file and verify it against the leaderboard cutoff.

2. Dataset Description

The dataset consists of individual heavy-equipment transaction records, split into a training set of 138,701 rows and a test set of 15,000 rows, with 50 and 49 columns respectively (test excludes the target). Table 1 summarizes the principal fields used in modeling.

Column	Description
TransactionID	Unique identifier for each transaction record
TargetValue (Target)	Transaction price in USD — right-skewed, range \$7,500–\$142,000
ManufactureYear	Year the equipment unit was originally produced
OperationalHoursMeter	Lifetime run-time hours logged by the machine
TransactionDate	Calendar date of the transaction — 3,676 unique values in train
Spec_FullDescriptor	Free-text string of technical parameters — 3,505 unique values
Spec_BaseClass / SubClass	Hierarchical equipment category — 1,249 / 147 unique values
RegionCode / VendorPartnerID	Geographic and vendor identifiers
col1 – col30	Anonymized technical specification fields (drivetrain, hydraulics, tires, etc.); several are missing in 80–99.95% of rows

Table 1. Principal dataset fields and descriptions.

An initial diagnostic pass — shape, dtypes, and missing-value profiling — confirmed the load matched the documented schema, with only 4 columns loading as native numeric dtype against 44 categorical/object columns. Missingness was heavily concentrated in the anonymized **colIN** specification fields: two columns (col18, col19) were missing in 99.95% of rows, with several further clusters missing in the 83–93% range. These clusters appear in matched pairs, consistent with optional equipment attachments that legitimately do not apply to every machine type, rather than random data loss.

3. Methodology

3.1 Exploratory Data Analysis

EDA covered the target distribution, missing-value patterns, numeric/categorical composition, and univariate and bivariate relationships against price. **TargetValue** was found to be right-skewed (skew = 0.97); applying

a log1p transform reduced this to near-symmetric (skew = -0.20) and aligns the training objective directly with RMSLE, which is mathematically equivalent to RMSE computed on log-transformed values. Scatter plots of manufacture year, meter hours, and asset scale against price showed wide, overlapping clouds rather than clean linear trends for any single feature, indicating that price is driven by interactions between age, usage, and equipment category rather than any one variable in isolation. Boxplots of price by utilization tier, equipment base class, and region showed clear median-price separation across categories, confirming that categorical fields — despite their high cardinality — carry substantial predictive signal.

3.2 Feature Engineering

Rather than feeding the two highest-cardinality raw fields directly into the model, each was reconstructed into a generalizable form before modeling, summarized in Table 2.

Feature	Description
EquipmentAge	Transaction year minus manufacture year, clipped at zero
HoursPerYear	Meter hours divided by equipment age — a usage-intensity signal
Txn_Year / Month / DayOfWeek	Calendar components decomposed from TransactionDate
Spec_TokenCount	Count of alphanumeric tokens in the free-text specification descriptor
Spec_FirstNumber	First numeric token parsed from the specification descriptor via regex
MissingSpecCount	Count of missing anonymized spec fields per row — turns a missingness pattern into a usable signal

Table 2. Engineered features used in modeling.

The same feature-construction function was applied identically to the training and test sets to avoid leakage and duplicated logic. The resulting feature matrix contained 55 usable columns — 12 numeric and 43 categorical, roughly tripling the numeric feature count from the original 4 — with informative missingness preserved via constant-category imputation rather than row deletion.

3.3 Model Training and Validation

Five candidate regressors were trained on an 80/20 train-validation split (110,960 / 27,741 rows), all fit on the log1p-transformed target and scored on RMSLE computed back on the original price scale: **Ridge Regression** ($\alpha = 5.0$) as a linear baseline; **Random Forest** (300 trees, max depth 16); **XGBoost** (600 trees, max depth 7, learning rate 0.05, histogram tree method); **LightGBM** (800 trees, 63 leaves, learning rate 0.03); and **CatBoost** (800 iterations, depth 8). All five shared a single preprocessing pipeline — median imputation and standardization for numeric features, constant-category imputation and ordinal encoding for categorical features — refit independently on the training fold for every model to prevent validation leakage. The best-performing model was then refined using **RandomizedSearchCV** (15 candidates, 3-fold cross-validation) over tree count, depth, learning rate, subsampling, and regularization strength.

4. Results and Model Performance

The five candidate models achieved the following validation performance, summarized in Table 3.

Model	RMSLE	MAE (USD)	R ²
Ridge (baseline)	0.5280	15,986.7	0.3779
Random Forest	0.2249	6,159.0	0.8872
LightGBM	0.2169	5,918.9	0.8950
CatBoost	0.2278	6,243.3	0.8842
XGBoost	0.2137	5,789.1	0.8981
XGBoost (tuned)	0.2066	5,546.3	0.9048

Table 3. Validation RMSLE, MAE, and R² by model.

Ridge Regression trailed the tree-based models by a wide margin (RMSLE more than double the weakest tree model), confirming that price in this dataset is governed by non-linear feature interactions rather than an additive linear combination of encoded fields. Among the tree-based models, the four candidates clustered within roughly 1.4 RMSLE points of one another, with **XGBoost** achieving the strongest untuned result. Randomized hyperparameter search over XGBoost's tree count, depth, learning rate, and regularization improved validation RMSLE from 0.2137 to **0.2066** — a relative gain of approximately 3.3% — with the search favoring a faster-converging configuration (learning rate 0.08, 500 trees, depth 8) over the hand-selected defaults. The tuned model was refit on the full 138,701-row training set and used to generate the final submission, which returned a leaderboard RMSLE of **0.20491** — closely matching the validation estimate, indicating the validation split was representative and the model was not overfit to a specific partition. This result sits marginally above the competition's 0.20 eligibility cutoff, with the near-fully-missing specification columns and untapped structure in the remaining anonymized fields identified as the most promising levers for closing the remaining gap.

5. Conclusion

This project demonstrates a complete, disciplined regression pipeline: the dataset was profiled for target skew, missingness structure, and categorical cardinality; high-cardinality raw fields were reconstructed into generalizable engineered features covering equipment age, usage intensity, and parsed specification content; and five candidate regressors were compared on a common validation split before the strongest model was refined via randomized hyperparameter search. The result — a tuned XGBoost model achieving validation RMSLE of 0.2066 and a leaderboard RMSLE of 0.20491 — represents a well-calibrated submission that closely tracks its own validation estimate, with a clearly identified path (denoising the near-empty specification columns, richer parsing of the remaining anonymized fields, and model blending) to close the small remaining gap to the competition's 0.20 cutoff.

6. Future Work

- Blend XGBoost, LightGBM, and CatBoost predictions (inverse-RMSLE weighted) rather than relying on a single final model.

- Drop or better-engineer the two columns missing in 99.95% of rows (col18, col19), which currently carry negligible signal.
- Parse numeric values out of the remaining anonymized colN text fields, converting more of the 44 categorical columns into genuinely numeric features.
- Pass CatBoost the original (unencoded) categorical columns directly via its native cat_features handling, rather than pre-encoding with OrdinalEncoder.
- Expand the randomized hyperparameter search (more iterations, wider grid) given additional compute budget.

Appendix — Technical Stack

The project is implemented in Python within a Kaggle notebook, using only the libraries permitted by the MLP project guidelines:

- pandas & numpy — data loading, cleaning, and feature computation
- matplotlib & seaborn — exploratory and diagnostic visualizations
- scikit-learn — imputation, scaling, pipeline construction, cross-validation, and evaluation metrics
- xgboost, lightgbm, catboost — gradient-boosted and CatBoost regression models
- re (regex) & scipy — specification-descriptor parsing and statistical utilities

Key output artifacts generated by the pipeline:

- submission.csv — final TargetValue predictions in the required submission format
- Target distribution, missing-value, univariate/bivariate, correlation, and feature-importance visualizations